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PROFESSIONAL SUMMARY

Computer Science undergraduate with a minor in Mathematics, skilled in Python, C++, and SQL. Experienced in software engineering across full-stack development and embedded systems, including firmware programming, hardware-software integration, and real-time control on Raspberry Pi and Arduino platforms. Strong analytical thinker with hands-on experience testing, debugging, and validating hardware and software systems.

EDUCATION

University of Massachusetts Lowell Lowell, MA

Bachelor of Science in Computer Science. Minor in Mathematics. GPA: 3.6. May 2027

Relevant Coursework: Software Engineering I, Data Structures, Discrete Mathematics (I-II), Probability and Statistics, Calculus (I-III)

Middlesex Community College Lowell, MA

Associate of Science in Computer Science. GPA: 3.6. December 2024

SKILLS

Programming Languages: Python, C++, SQL, JavaScript, HTML, CSS, Flask

Tools and IDEs: VS Code, IntelliJ, GitHub, Jupyter Notebook, Arduino IDE, Git

Operating Systems: Linux (Ubuntu), Windows

PROJECTS

Picar Robot (Python, Flask, Raspberry Pi, Linux, HTML, CSS, JavaScript, LLM, Gemini API)

- Built a voice assistant integrating speech-to-text, the Gemini LLM API, and text-to-speech to enable the robot to answer spoken questions
- Developed a Flask REST API and responsive web controller for real-time driving and camera control over Wi-Fi, with live MJPEG video streaming
- Programmed autonomous obstacle avoidance on a Raspberry Pi Zero 2 W using live ultrasonic sensor data to detect and steer around obstacles
- Debugged memory-constrained Linux over SSH, resolving GPIO conflicts, audio-device routing, and process lifecycle issues

File Finder - Desktop Search Application (Python, PySide6, SQLite, FTS5, PyMuPDF)

- Built a desktop app that lets users find files by searching for words inside them, instead of remembering file names or locations.
- Created a system to scan and index TXT, PDF, and DOCX files, using SQLite for fast searching.
- Built a responsive desktop interface with PySide6 and packaged the project as a standalone macOS application.

PingPong Game (Arduino UNO R4 WiFi C++, Arduino IDE, LED Matrix, Embedded Systems)

- Developed an interactive two-button Pong game on a 12x8 LED matrix with real-time ball physics and collision detection
- Implemented responsive button controls (pins 13, 12) for paddle movement with frame-counting for independent ball and player speeds
- Integrated ArduinoGraphics library for text rendering and intro/game-over screen displays

Sokoban Game (C++, SFML, Boost)

- Built a fully playable graphical puzzle game with object-oriented design, using a class inheriting from SFML's Drawable interface
- Implemented keyboard controls, undo/redo with move-history stacks, and level loading via overloaded stream operators
- Wrote 10 Boost unit tests covering movement, win conditions, and edge cases

EXPERIENCE

Test Operator (ICT/FCT) February 2026 – Present

Mack Technology Westford, MA

- Flashed firmware onto PCBs by executing programming scripts
- Tested and validated PCBs using In-Circuit (ICT) and Functional (FCT) testing methodologies
- Debugged hardware failures by analyzing pass/fail test outputs
- Collaborated with engineers to diagnose defects and retest systems